

Properties of sodium-based ternary phosphate glasses produced from readily available phosphate salts

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Abstract

Three series of phosphate glasses were produced by melting together sodium phosphate salt (NaH_2PO_4) and the phosphate salts of either calcium (CaHPO_4), magnesium ($\text{MgHPO}_4 \cdot 3\text{H}_2\text{O}$) or iron ($\text{FePO}_4 \cdot 2\text{H}_2\text{O}$) in a 5% gold/95% platinum crucible at 1200 °C. The glass compositions were confirmed by EDX and XRD analysis. Glass transition temperature (T_g), density and durability in water were determined for all the compositions. Maximum metal oxide contents before devitrification were between 55% and 59% for $\text{CaO} + \text{Na}_2\text{O}$ and 59% and 62% for $\text{MgO} + \text{Na}_2\text{O}$. The normalized equivalent for $\text{Fe}_2\text{O}_3 + \text{Na}_2\text{O}$ was between 55% and 61%. Density values for the glasses lay between 2.49 and 2.75 g cm^{-3} . T_g s lay between 295 °C and 470 °C. Degradation rates in deionized water at 37 °C lay between 0.03 $\text{g cm}^{-2} \text{h}^{-1}$ for Na phosphate glasses and $9 \times 10^{-6} \text{ g cm}^{-2} \text{h}^{-1}$ for Ca phosphate glasses, $3 \times 10^{-6} \text{ g cm}^{-2} \text{h}^{-1}$ for Mg phosphate glasses and $<3 \times 10^{-6} \text{ g cm}^{-2} \text{h}^{-1}$ for Fe phosphate glasses. The effect of metal addition on properties goes as $\text{Fe} > \text{Mg} > \text{Ca}$ for degradation rates and T_g and $\text{Fe} > \text{Mg} \approx \text{Ca}$ for density. The change in properties with metal addition was seen to be linear for Fe and Ca additions but not with Mg addition. This is in agreement with the anomalous behavior of magnesium phosphate glasses.

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1. Introduction

The increase in durability of phosphate glass formulations has allowed them to achieve practical application. In fact, phosphate glass has been shown to have highly desirable properties in a number of areas given its variable degradation rate, UV transparency and high thermal expansion coefficient. This has led to application in optics [1,2], solid state lasers [3], dosimeters [4], high temperature seals [5] and nuclear waste containment [3]. Amongst the improved properties are the mechanical characteristics of the glass, with it being possible to produce phosphate

glasses that exhibit mechanical properties that are on a par with silica glasses [6].

Phosphate glasses are inorganic polymers based on the tetrahedral phosphate anion, with charges on the polyanions balanced by the presence of metal cations [7]. Unfortunately, limited practical use was seen for these glassy materials in the early years of their study since they were highly susceptible to hydrolytic attack. Silica glasses did not suffer this drawback [8] and so phosphate glass did not appear to offer itself as an alternative material. However, relatively recently, it was discovered that the durability of phosphate glasses could be greatly increased, with it being possible to control the degradation rate of the glass by altering its composition both in terms of P_2O_5 content and cation species [3,9,10].

As a general rule, metal cations of higher valency are capable of creating more metal–oxygen cross-links within

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the polymeric structure of phosphate glass, which make the glass more durable. Cations with greater field strength increase the covalent character of bonds within the glass and increase the glass durability [3,9]. Variation in P_2O_5 content determines the overall length of polyphosphate chains in the glass as well as the extent of phosphate cross-linking, though the effect of this on the physical properties of the glass has been shown to be considerably less significant than changing the metal ion [3]. Phosphate glasses have been produced that exhibit degradation rates of between a few hours and many years [11,12].

It is possible to produce fibres from phosphate glasses and indeed, studies in the US have shown this process to be considerably less energy intensive (up to 60% lower energy requirements) than the production of E-glass fibre [13,14]. This opens up the possibility of the use of phosphate glass as a replacement material for E-glass for use in composite reinforcement, potentially a huge new market for phosphate glasses. In addition to durable composite materials, the ability to vary the degradation rate of phosphate glasses provides great potential in biodegradable composites for medical application and recyclable/composable materials.

Since the properties of a phosphate glass are highly dependent on the nature and amount of additives in the glass, they will vary greatly between different glass compositions. In order to be able to effectively use phosphate glasses, it is necessary to be able to make reasonable predictions of the properties of the glass as the composition changes. It is the intent of this paper to determine some simple relationships between glass composition and bulk properties.

In this study, sodium phosphate-based glasses were produced from phosphate salts whereby a proportion of the Na^+ was replaced with Ca^{2+} , Mg^{2+} or Fe^{3+} . This provides a comparison of cations of different valency and of different field strength. The salts chosen also allow for the variation in phosphate content without the use of P_2O_5 (Fig. 1). The glasses were characterized in terms of feed and final composition, density, glass transition temperature and degradation rate in deionized water. These properties were then plotted as a function of composition and relationships determined.

The phosphate salts that were chosen for use in this study were selected for their ready availability, ease of han-

dling and low hazard in comparison to the required oxides: although iron oxide is considered benign, magnesium, calcium and sodium oxides are rated as hazardous. The use of phosphorus pentoxide would also present chemical hazards during production and its hygroscopic nature would make open weighing of precursor materials difficult.

2. Materials and methods

Sodium dihydrogen phosphate (NaH_2PO_4), calcium hydrogen phosphate ($CaHPO_4$), magnesium hydrogen phosphate trihydrate ($MgHPO_4 \cdot 3H_2O$) and iron(III)phosphate dihydrate ($FePO_4 \cdot 2H_2O$) were used as received (Sigma–Aldrich). Appropriate amounts of each salt (as shown in Table 1) were mixed together for 30 min using a Turbula powder mixer. Production of glasses with larger quantities of calcium, iron and magnesium was attempted but they could not be cast without crystallisation occurring.

The phosphate salts were placed into Pt/Au (5%Au/95%Pt – Johnson Matthey) crucibles. The samples were dehydrated for 1 h at 350 °C before being transferred to a Carbolite muffle furnace at 1200 °C. After 4 h the samples were removed from the furnace, swirled to homogenize the melt and then cast onto a polished steel plate that was at room temperature. The glass was then allowed to cool in air.

Once cool, portions of each sample were crushed to powder and the glass transition temperature (T_g) determined using Differential Scanning Calorimetry (DSC – Perkin–Elmer DSC 7; performed under nitrogen with a heating rate of 10 °C min⁻¹). Portions of each of the glass samples were cast into epoxy resin and then polished with SiC paper and diamond cloths, using industrial methylated spirit (IMS) as an eluent. The samples were then dried and cleaned with a dry air spray before being carbon coated. These samples then underwent energy dispersive X-ray (EDX – Oxford Instruments INCA X-sight linked to a Philips XL-30 SEM) analysis. A standards-based ZAF program with primary reference standards and profiles was used for quantification. Archimedes density measurements were performed on small samples of the glasses using a perfluoro(tetradecahydrophenanthrene) oil (Fluorochem PP11).

The remainders of the glass samples were then re-melted at 1200 °C and cast into graphite moulds at ≈ 20 °C above the T_g of the glass. The glass was held at this temperature for 90 min before being cooled at 1/4 °C min⁻¹ down to 100 °C below the T_g . The furnace was then switched off and allowed to cool. For glasses with a T_g above 450 °C it was not possible to anneal in the graphite mould. These samples were cast into the graphite at 450 °C and cooled slowly to prevent cracking. Once cool they were removed from the mould, put back into the furnace and annealed at the appropriate temperature as with the other glasses.

The glass rods so obtained were 100 mm in length and 10 mm in diameter. These were sectioned into 2 mm and

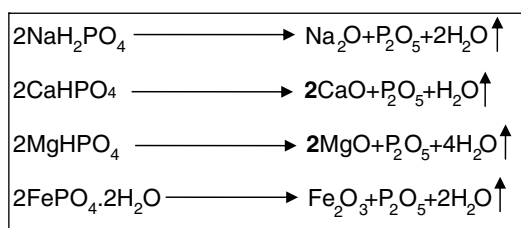


Fig. 1. Diagram showing the reaction scheme for each of the phosphate salt precursors. The sodium and iron salts provide one mole of oxide for every two moles of salt. The calcium and magnesium salts provide one mole of oxide for each mole of salt.

Table 1
Glass compositions

Glass code	Mole percent		Mass percent	
Fe Glasses	NaH ₂ PO ₄	FePO ₄ · 2H ₂ O	NaH ₂ PO ₄	FePO ₄ · 2H ₂ O
1 Fe	99	1	98.45	1.55
3 Fe	97	3	95.40	4.60
5 Fe	95	5	92.42	7.58
10 Fe	90	10	85.25	14.75
15 Fe	85	15	78.45	21.55
20 Fe	80	20	71.99	28.01
Ca Glasses	NaH ₂ PO ₄	CaHPO ₄	NaH ₂ PO ₄	CaHPO ₄
10 Ca	90	10	88.81	11.19
20 Ca	80	20	77.92	22.08
30 Ca	70	30	67.32	32.68
40 Ca	60	40	56.90	43.10
Mg Glasses	NaH ₂ PO ₄	MgHPO ₄ · 3H ₂ O	NaH ₂ PO ₄	MgHPO ₄ · 3H ₂ O
10 Mg	90	10	86.09	13.91
20 Mg	80	20	73.33	26.67
30 Mg	70	30	61.69	38.31
40 Mg	60	40	50.74	49.26
50 Mg	50	50	40.83	59.17

n.b.: The glass codes used to identify the different formulations in this study refer to the total content of metal in the glass, not metal oxide.

10 mm lengths for X-ray diffraction (XRD – Siemens D500) and degradation testing, respectively. The samples for degradation testing were weighed and their dimensions measured in triplicate using a micrometer. They were then placed into sealed glass vials with 25 ml of deionized water and put into a re-circulating waterbath set at 37 °C. This temperature was used as there is interest in using the materials in biomedical application. The glasses were removed from the water and weighed (in grams, to 4 decimal places) and measured (in mm, to 3 decimal places) at regular intervals over the course of six weeks. This method was suitable to measure glass degradation down to about $10^{-6} \text{ g cm}^{-2} \text{ h}^{-1}$.

3. Results

The experiments provided maximum thresholds for the substitution of iron, calcium and magnesium ions into a sodium phosphate whilst retaining an amorphous glass structure. For Fe₂O₃ this is between 10% and 15%, for CaO it lies between 33% and 40% and for MgO it lies between 40% and 46% (information is not included on them in this paper but attempts were made to produce glasses with 15% Fe₂O₃, 40% CaO and 46% MgO, all of which crystallized). All the results for the glasses that were produced successfully are summarized in Tables 2 and 3. Correlation of feed and final (EDX derived) compositions is plotted in Fig. 2; it can be seen that there is a good agreement between nominal and measured compositions.

Fig. 3 shows values of degradation rate plotted against P₂O₅ content. There is no clear trend between P₂O₅ content and degradation rate and no clear trend was found for glass transition temperature or density. Figs. 4–6 show values of density, glass transition temperature and degrada-

Table 2
Physical properties of the glasses

Glass	T _g (°C)	Density (g cm ⁻³)	Degradation rate (g cm ⁻² h ⁻¹)
0 Fe	295	2.49	0.3 ^a
1 Fe	302	2.51	0.19996
3 Fe	309	2.51	0.10261
5 Fe	314	2.54	2.587E-02
10 Fe	333	2.61	3.0E-04
15 Fe	354	2.66	1.E-06
20 Fe	384	2.75	3.E-06
0 Ca	295	2.49	0.3 ^a
10 Ca	310	2.54	0.05025
20 Ca	331	2.59	5.43E-03
30 Ca	350	2.62	9.E-05
40 Ca	390	2.67	9.E-06
0 Mg	295	2.49	0.3 ^a
10 Mg	332	2.53	8.75E-03
20 Mg	367	2.58	2.E-05
30 Mg	403	2.62	9.E-06
40 Mg	425	2.67	6.E-06
50 Mg	470	2.67	3.E-06

^a Sodium glass dissolves very quickly and an accurate determination of rate was not possible.

tion rate plotted against metal oxide ratios for calcium, magnesium and iron. Good correlation is exhibited between the metal content and the density, the glass transition temperature and the degradation rate.

Measurements of glass degradation were taken over regular intervals and used to calculate the degradation rate at each time point, in order to ensure that the degradation rate was linear over the course of the experiments. The final rates stated in Table 2 are then an average of all of these measured rates. It should be noted that these degradation rates are specific to a static system and the particular media

Table 3
Feed and EDX derived compositions of the glass formulations

Glass	P ₂ O ₅ feed	P ₂ O ₅ EDX	Na ₂ O _{feed}	Na ₂ O _{EDX}	Fe ₂ O ₃ feed	Fe ₂ O ₃ EDX	Fe ₂ /(Fe ₂ + Na ₂)	Total MO content (Fe + Na) ^a		O/P ratio	
0 Fe	50	53	50	47	0	0	0	47	47	47	3.0
1 Fe	50	53	50	46	1	0	0.01	47	48	47	3.01
3 Fe	50	54	49	45	2	1	0.03	47	49	47	3.03
5 Fe	50	53	48	44	3	2	0.05	48	51	49	3.05
10 Fe	50	53	45	42	5	5	0.10	50	57	51	3.1
15 Fe	50	52	43	41	8	7	0.15	51	62	54	3.15
20 Fe	50	52	40	39	10	9	0.19	52	66	55	3.2
					CaO _{feed}	CaO _{EDX}	Ca/(Ca + Na ₂)	(Ca + Na ₂)			
0 Ca	50	53	50	47	0	0	0	47			3.0
10 Ca	48	53	43	38	10	10	0.20	47			3.05
20 Ca	46	49	36	32	18	19	0.38	51			3.1
30 Ca	44	47	30	26	26	27	0.51	53			3.15
40 Ca	42	44	25	22	33	33	0.60	56			3.2
					MgO _{feed}	MgO _{EDX}	Mg/(Mg + Na ₂)	(Mg + Na ₂)	For Fe		
0 Mg	50	53	50	47	0	0	0	47	Column 1 is Fe ²⁺		3.0
10 Mg	48	50	43	40	10	10	0.21	50	Column 2 is Fe ³⁺		3.05
20 Mg	46	48	36	33	18	19	0.36	52	Column 3 is		3.1
									20%Fe ²⁺ /80%Fe ³⁺		
30 Mg	44	46	30	28	26	27	0.49	54			3.15
40 Mg	42	44	25	22	33	34	0.60	56			3.2
50 Mg	40	42	20	19	40	40	0.68	58			3.25

^a M = divalent cation.

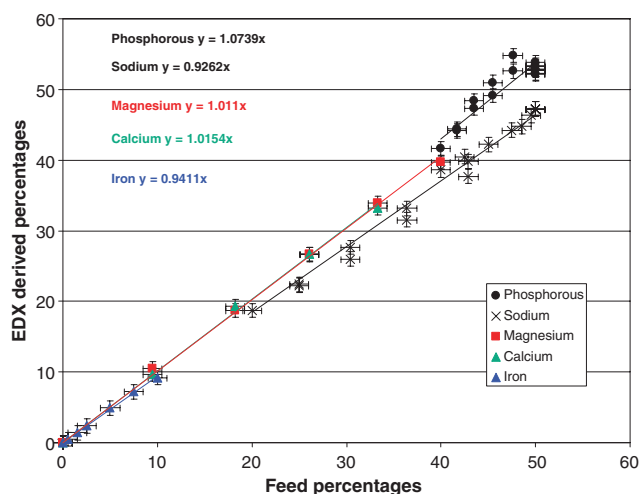


Fig. 2. Graph showing the correlation between feed and EDX-derived final compositions of the glasses.

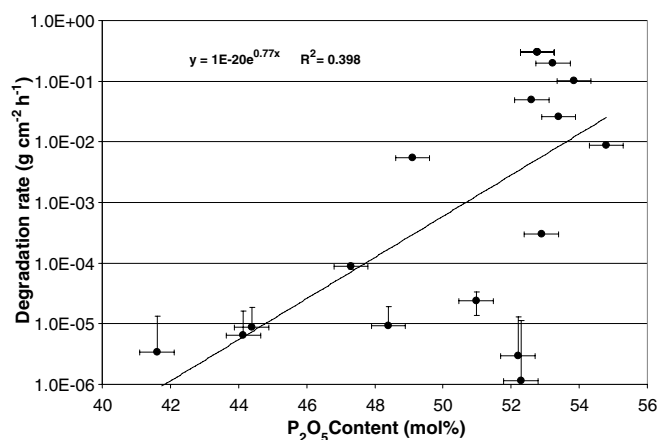


Fig. 3. Graph showing the variation in degradation rate of the EDX-derived glass compositions with respect to P_2O_5 composition.

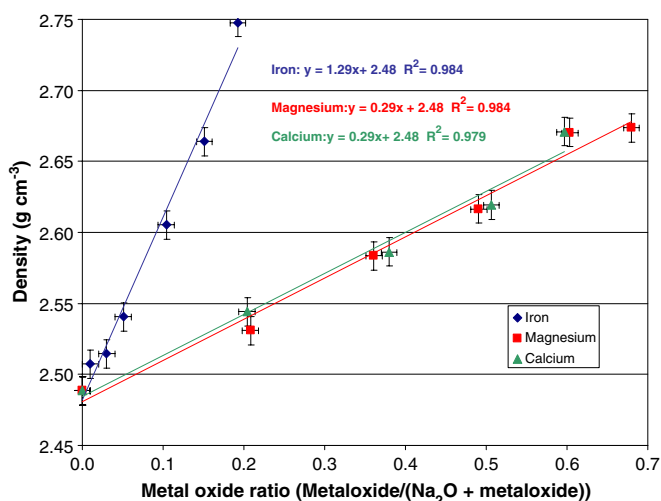


Fig. 4. Graph showing the variation in density of the glass compositions with respect to increasing metal oxide content.

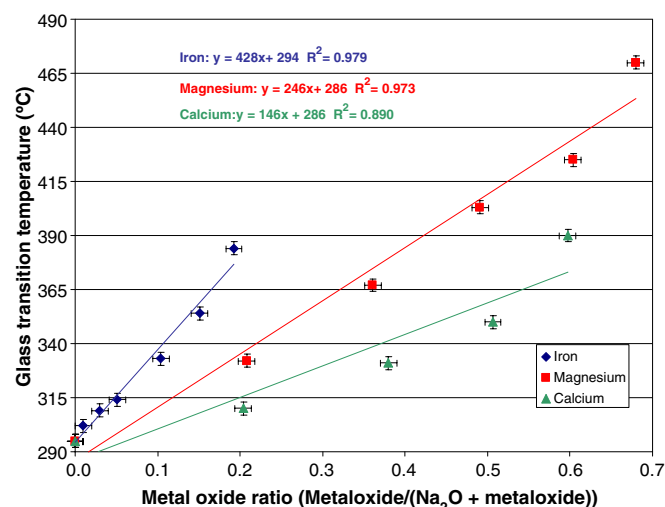


Fig. 5. Graph showing the variation in glass transition temperature of the glass compositions with respect to increasing metal oxide content.

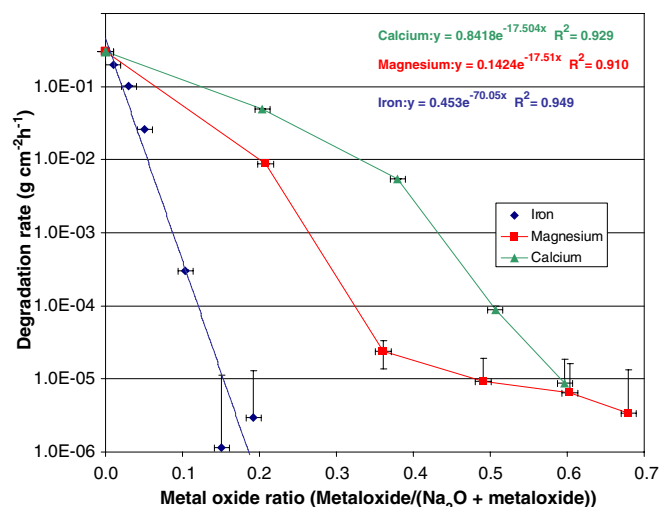


Fig. 6. Graph showing the variation in degradation rate of the glasses with respect to increasing metal oxide content.

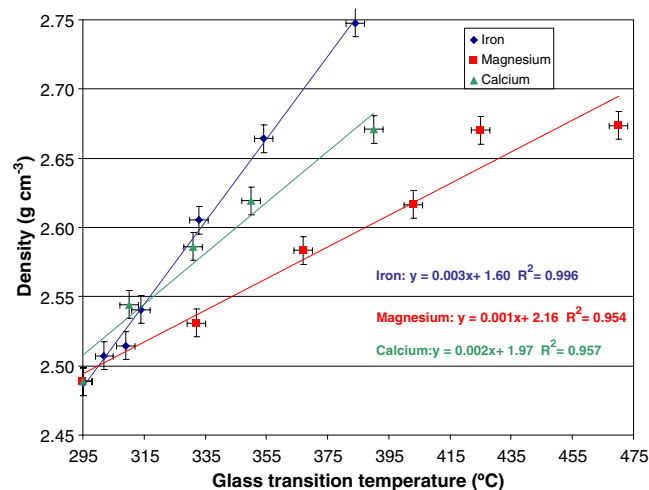


Fig. 7. Graph showing the variation in density of the glass compositions with respect to their glass transition temperatures.

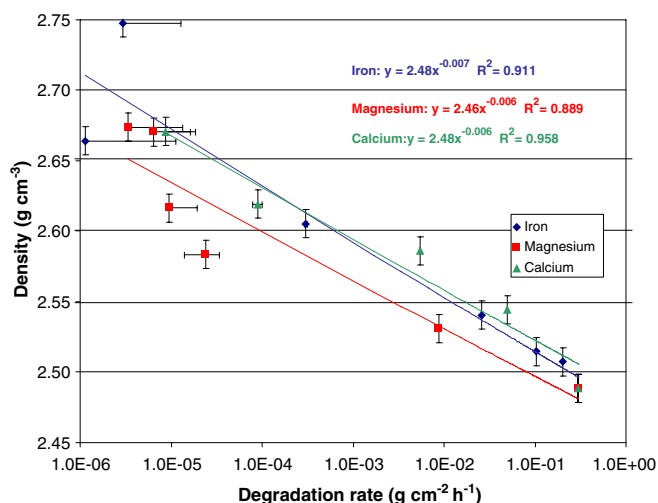


Fig. 8. Graph showing the variation in density of the glass compositions with respect to their degradation rates.

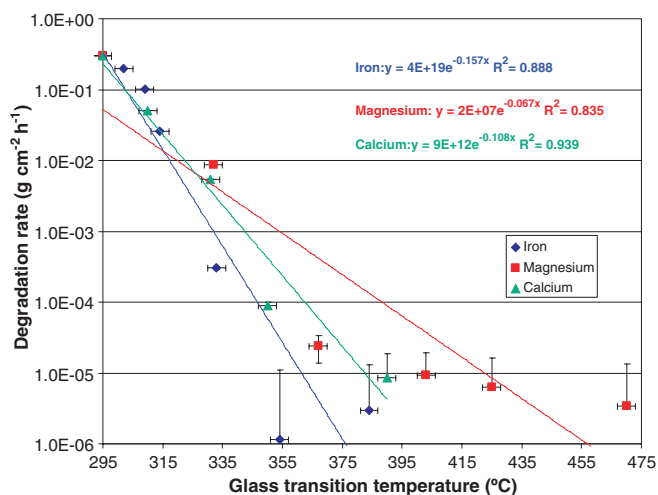


Fig. 9. Graph showing the variation in degradation rates of the glass compositions with respect to their glass transition temperatures.

used. If a continuous flow through of degradation media were used, the values could be significantly different.

Overall, an increase in T_g and density is observed with increasing metal oxide ratio, whereas degradation rate decreases in a non-linear fashion (Fig. 6). Figs. 7–9 compare the values of density, T_g and degradation rate with each other. Density and T_g show a good linear correlation, whereas degradation rate does not provide a good correlation with either T_g or density.

4. Discussion

The final glass compositions determined by EDX analysis match quite closely to the calculated feed values for the included metal oxides (Table 3, Fig. 2). There is no evidence of gold or platinum within the glasses that would indicate contamination of the glass by the crucible in which they are made. When mixing a glass from basic oxides one

might expect that some P_2O_5 [15] and potentially some Na_2O [15–17] could be lost through volatilisation (albeit only a small amount). However, in this case where the glasses are produced from phosphate salts no P_2O_5 or Na_2O are formed that would be able to boil off. The P_2O_5 content is universally higher than expectation, by anywhere from 4% to 15%, whereas the Na_2O content is always lower than expected (by 5–14%).

EDX is considered to be accurate to about 1%, so this is insufficient to be responsible for the discrepancy. However, EDX can have difficulty in detecting light elements and cannot directly measure water content. The fact that the higher value of P_2O_5 is almost matched by the lower value of Na_2O may be indicative of this difficulty in measurement but this is unlikely. Care was taken to ensure that the beam was not too intense (to reduce the likelihood of element migration) and primary reference standards were used. The other possibility is that the precursor material composition varies from what is expected. Since the calcium, magnesium and iron values are very close to expectation this suggests that the sodium dihydrogen phosphate may be the source of the variation. EDX was performed on the NaH_2PO_4 powder to check its composition. The expected composition is 50% P_2O_5 and 50% Na_2O , however, the values obtained were 52.7% P_2O_5 and 47.3% Na_2O . These values are very close to those observed for the glass that was produced using only this precursor (0 Fe, 0 Ca and 0 Mg, Table 3). It is surmised that the deviation from expected values in the glasses are due to precursor composition rather than loss of material on melting.

Some values for maximum ion incorporation in phosphate glasses are provided by Rawson [18] with maximums for calcium and magnesium in binary phosphate glasses given as between 56% and 58% for CaO and 60% and 65% for MgO. Additionally, maximum values for Na_2O are listed as 60% but none are provided for Fe_2O_3 . Uo et al. [19] provide a phase diagram for a ternary phosphate glass containing calcium and sodium, showing an upper limit for CaO of around 60% also. Interestingly, if you consider total ion content for the glasses in this study ($Na_2O + CaO$, $MgO + Na_2O$, etc.) as shown in Table 3, maxima for the CaO and MgO glasses lie between 55% and 59% and 59% and 62%, respectively. This is very close to the values given by Rawson for the CaO and slightly low for the MgO – though this may be due to the limit of 60% for Na_2O , with the Na_2O content lowering the overall maximum oxide content.

This does not follow for the iron phosphate glasses, with all the values of $Na_2O + Fe_2O_3$ lying between 46% and 47%. This is because Fe_2O_3 is not directly comparable to CaO, MgO or Na_2O . Within a phosphate glass, each phosphorus atom has an associated O^- , to which the metal ions can co-ordinate. Ca^{2+} and Mg^{2+} will co-ordinate to two of these sites and Na^+ will co-ordinate to 1. Thus, CaO, MgO and Na_2O will each be expected to account for 2 sites [7] and moles of each are directly comparable. This is not the case for Fe_2O_3 as it contains 2 Fe^{3+} ions, which would

account for 6 sites in the glass. Additionally, despite the feed of Fe ions being in the 3+ state, they can be reduced during glass melting to the 2+. This would mean an initial loading of Fe_2O_3 could, after melting, actually only take up 4 sites. In order to make the values for iron directly comparable to CaO and MgO the total ion content can be adjusted and normalized by assuming a 2+ or 3+ oxidation state (see Table 3). This sees an effective maximum for the Fe glass of between 52% and 56% (if the iron is in the 2+ state) and between 63% and 68% (if the iron is in the 3+ state).

Karabulut et al. [20] performed Mossbauer and X-ray absorption fine structure spectroscopy to determine the ratio of Fe^{2+} to Fe^{3+} in iron phosphate glasses. Although these glasses are not of the same composition as those in this study it provides a useful estimate. At 1200 °C in air the proportion of Fe^{2+} was 20%. If we incorporate this into the figures for the iron phosphate glass in this study, it gives a maximum effective value of $\text{Fe}_2\text{O}_3 + \text{Na}_2\text{O}$ of between 55% and 61%. These results would indicate that the ease of incorporating ions into phosphate glass goes in the order $\text{Mg} > \text{Fe} > \text{Ca}$, under these conditions of glass manufacture.

As well as considering the overall metal ion content, the oxygen/phosphorous ratio of the glasses can also be considered (Table 3). O/P ratio is an indication of the phosphate structure within the glass; with an O/P ratio of 3 indicating theoretically infinite phosphate chains and values above this indicating progressively shorter chains (splitting a phosphate chain requires the inclusion of an extra oxygen). It can be seen in Table 3 that for each of the glass types, the highest metal ion content glass has an O/P ratio of around 3.2. This may be a general indicator of glass formation limit and warrants future investigation.

The correlation between metal content and physical properties is readily apparent (Figs. 4–6). Values for density show a linear correlation (Fig. 4), with all the metals providing co-efficients greater than 0.95. The correlation for T_g is almost as good (Fig. 5), though interestingly the last point for each of the glasses is significantly higher than expected (this can also be seen in Fig. 6 but is less pronounced). Previous work [21] had seen a drop in T_g at higher levels of metal incorporation and this was believed to be caused by a disruption of the P–O–P chains and the presence of nano-particles of metal oxide. There is no obvious reason for an increase to occur, there is no evidence to suggest phase separation in the DSC analysis and the trend occurs whether the phosphorous content is maintained (Fe phosphate glass) or varied (Ca and Mg phosphate glasses). It may suggest that at these levels there are sufficient amounts of the metals present for them to begin to interact, with the formation of M–O–M bonds as a precursor to phase separation and crystallisation that occurs at slightly higher metal content. This would suggest that the maximum inclusion of metal in the glass that can still provide predictable properties is lower than the absolute maximum amount of metal that can be incorporated.

The correlation of the degradation rates of the glasses is less clear (Fig. 6), the iron phosphate glass follows a relatively linear path but a simple relationship seems unsuitable for calcium and magnesium. It could be argued that this is related to the change in phosphorous content with increasing metal but there was no obvious relationship in Fig. 3 and the normally close values of magnesium and calcium significantly diverge in Fig. 6: the degradation rate for the magnesium phosphate glass initially decreases rapidly and then levels off, whilst the inverse is true for the calcium glasses, for which the degradation rate drops off more rapidly at higher concentrations. This effect is surprising and suggests that magnesium has a more profound effect on the degradation rate at lower concentrations. At maximum inclusions of magnesium and calcium the degradation rates are very similar.

This may be related to the reported ‘anomalous’ behavior of magnesium in phosphate glasses, which is believed to switch from a four co-ordinate to a six co-ordinate structure at a composition of 50% P_2O_5 [22]. In Fig. 6 this transition of co-ordination states would occur for magnesium glasses between the third and fourth points (around about 0.4–0.45 on the x -axis) and indeed, a two step graph could be envisioned. Sahar and Kamaruddin reported the formation of a magnesium pyrophosphate structure in similar glasses, which provided an increase in stability [23]. However, in CaO glasses there was no apparent discontinuity in glass durability and no calcium pyrophosphate was observed. Interestingly, the authors also reported an increase in density for Ca containing glasses but a decrease for Mg glasses. This contradicts the results seen here where the density increases for both Ca and Mg glasses.

Others dispute this change in co-ordinate number, however, citing results that do not show any significant variation but instead suggest a mixture of co-ordination states [24]. Their rationalisation of the difference is that as the glass moves to a composition of $\text{P}_2\text{O}_5 < 50\%$ the length of the phosphate chains reduces, causing a reduction of steric hindrance in the structure and an associated increase in packing density. This increase in packing density could potentially account for the sudden increase in durability since the rate of water diffusion into the structure would be reduced. It would also cause an increase in macroscopic density. The paper does not consider calcium glasses. Furthermore, Walter et al. observed heterogeneities of around 1 nm in size in Mg phosphate glasses that have a 45.5% or greater MgO content. This is quite interesting, since in this study, attempts were made to produce a 60 Mg glass but it crystallized: this glass would have contained 46% MgO. This gives some credence to the suggestion that the cause of the unexpected T_g increase (Fig. 5) is related to the development of small, metal rich sites prior to full crystallisation. The T_g increase is seen at 40% Mg in the glasses in this study, which is significantly lower than their 45.5% threshold but the glasses in this study also contain sodium. While sodium does not create cross-linking in the structure that would lead to an increase in glass transition

temperature, it does take up positions on the phosphate chain that could otherwise be occupied by magnesium and thus the threshold at which magnesium will begin to phase separate is lower. Calcium glasses achieve a 'good' level of durability [25] at 40% inclusion, magnesium glasses at 30% and iron glasses at 15%.

Fig. 3 shows the phosphate content of all the compositions, regardless of the metal ion content, and the effect on density. There is little correlation. Figs. 4–6 show metal ion contents, regardless of phosphate levels. In these cases there is a significant correlation. Clearly the metal oxide content is the over-riding factor affecting glass properties, with phosphate content showing little effect. This apparent lack of dependence of glass properties on phosphate supports the reports of Brow [3] and observations in many other papers [6,10,19,25–27].

Figs. 7–9 are graphs correlating the physical properties of the glasses with each other. A good correlation between these values would allow prediction of glass properties by measurement of just one physical property. The linear correlation between T_g and density is good, with Fe, Ca and Mg all having correlation co-efficients of above 0.95. The correlation of degradation rate is less good. Calcium has the best correlation, followed by iron and then magnesium.

The correlation for the calcium phosphate glass is good enough to suggest that the perceived curve in the degradation graph (Fig. 6) is not significant. The discrepancy for magnesium is related to the anomalous properties of the glass as previously discussed. The discrepancy for the iron glasses is most likely due to the values of degradation rate for the high iron content glasses being close to the detection limit for the experiment. The true degradation rate for the iron phosphate glasses is likely to be $<3 \times 10^{-6} \text{ g cm}^{-2} \text{ h}^{-1}$. It can be seen in both Figs. 8 and 9 that the four lower iron content glasses give a very good relationship. It is expected that iron and calcium phosphate glasses have properties that are easily predictable, whilst magnesium phosphate glasses do not.

Because of the wide variation in the glass compositions that have been studied it is difficult to make direct comparisons to other published results. However, there are a few examples that are useful. Rajaram and Day [28] produced both magnesium and calcium doped sodium phosphate glasses. Their 9.4% magnesium phosphate glass is similar in composition to the 10 Mg glass in this study, with these having T_g s of 335 °C and 332 °C, respectively. Their 9% calcium glass is also similar to the 10 Ca glass in this study, with both having identical T_g s of 310 °C. In both instances Rajaram's glasses contain less P_2O_5 but the relatively close values suggest the minor effect the phosphate chain has in comparison to the divalent metal content. Rajaram's degradation rates were markedly lower (by an order of magnitude), though they were performed at 30 °C rather than 37 °C. This would suggest a significant temperature dependence of the degradation rate.

Ahmed et al. [29] also produced phosphate glasses containing both sodium and calcium. The 40 Ca glass in this

study falls neatly between his $\text{P}_{45}\text{C}_{30}\text{N}_{25}$ (i.e., $\text{P}_{45}\text{Ca}_{30}\text{Na}_{25}$) and $\text{P}_{45}\text{C}_{35}\text{N}_{20}$ formulations, as does the T_g value: 40 Ca has a T_g of 390 °C, almost exactly halfway between his values of 375 °C and 407 °C. Also, the degradation rates are of the same order of magnitude (with the degradation study being performed under the same conditions).

Furthermore, Franks et al. [30] produce a series of phosphate glasses containing sodium and calcium. Her $\text{Ca}_{32}\text{Na}_{23}\text{P}_{45}$ composition is very close to the 40 Ca glass in this study; it provides a T_g of around 380/390 °C in comparison to 390 °C.

No suitable examples were found for the comparison of the properties of phosphate glasses containing iron.

The closeness of these values gives a good indication that resultant glasses behave similarly regardless of the precursors used in their production (variously anhydrous and hydrous phosphate salts, carbonates and hydroxides). It also provides confidence in the results presented here and a good expectation that a predictive model can be developed for phosphate glass properties. However there is one paper that does not find agreement with these results: Uo et al. [19] produce a number of calcium glasses. The 10C50P and 10C60P glasses lie around the composition of the 20 Ca glass in this study and the 20C60P and 20C50P glasses lie around the 40 Ca composition. In both cases the glasses produced by Uo have much lower T_g values (190–219 °C cf. 331 °C and 200–256 °C cf. 390 °C). However, Uo used ammonium salts and did not perform any compositional analysis. There is no obvious reason why these results are so much lower, although the glasses are produced at a lower temperature for a shorter amount of time. This may have meant that formation of the glass was not fully complete before it was quenched [21] and also that significant amounts of residual water may have remained in the glass.

A phosphate glass for optics use is considered to have a 'good' durability if it has a degradation rate of around $10^{-5} \text{ g cm}^{-2} \text{ h}^{-1}$ or less [25]. However, this is insufficient for use as a medical composite reinforcement where the glass would be continuously exposed to an aqueous environment. A nominal 1 m length of 15 μm diameter glass fibre of the 20 Fe composition would weigh $\approx 4.9 \times 10^{-4} \text{ g}$ and have a surface area of 0.47 cm^2 . In order to have lost 100% of its mass in 6 weeks it would need a degradation rate of at least $1 \times 10^{-6} \text{ g cm}^{-2} \text{ h}^{-1}$, however, the intention is to maintain structural support throughout the six weeks and so a mass loss of 20% would be more acceptable. This would require a degradation rate of around $2 \times 10^{-7} \text{ g cm}^{-2} \text{ h}^{-1}$ or better, an order of magnitude lower than that observed for any of the ternary formulations in this study.

In order to produce a glass useful for structural reinforcement of medical composites (or indeed the reinforcement of any composite material that would be exposed to a wet environment) it may be necessary to combine different di- and tri-valent metals together to form quaternary or higher mixtures at the expense of sodium. This would require either the use of different precursor salts to those

described here (perhaps $\text{Ca}(\text{H}_2\text{PO}_4)_2$ or similar) or the inclusion of phosphorus pentoxide since CaHPO_4 and MgHPO_4 alone will not melt to form glasses.

5. Conclusions

Phosphate glasses can be produced from phosphate salt precursors with closely matching feed and final compositions. There is a good correlation between the di- and tri-valent metal content of the glasses and their properties but there is no significant correlation between the phosphate content and the glass properties. Thus the cross-linking of the glass structure by di- and tri-valent metals has a more significant (over-riding) effect on properties than the state of the phosphate chain backbone. Density and glass transition temperature increase and degradation rates go down with increasing di and tri-valent metal content in all cases. Maximum contents before crystallisation in the glasses are between 55% and 59% for CaO , between 59% and 62% for MgO and between 55% and 61% for an assumed mix of 20% Fe_2O_3 and 80% Fe_2O_3 . The effect of cation content has the order $\text{Fe} > \text{Mg} > \text{Ca}$ for T_g and degradation rate and $\text{Fe} > \text{Mg} \approx \text{Ca}$ for density. The glass properties should be predictable for Fe and Ca containing glasses (except near the crystallisation limit) but not Mg glasses. Knowing one physical property could be used to predict other physical properties in Fe and Ca containing glasses, but not Mg glasses.

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